INSTRUCTIONS

Motivation: Based on human classification of the tidal features and low-surface-brightness components around galaxies using LSST-like mock images, our aim is to understand:

- The robustness of such classifications
- If / how classifications vary as a function of galaxy orientation, surface-brightness limit and redshift.
- What range of surface-brightnesses certain tidal features are typically visible at and what surface-brightness LSST will need to reach for reliable classifications to be made.

Our analysis of these classifications will form a significant part a paper that will published as part of a series of papers by the LSB working group of the LSST science collaboration.

Mock images: Mock images have been generated from a number of objects from the NewHorizon simulation ($10^{9.5} M_{\odot} < M_{\star} < 10^{11.5} M_{\odot}$) for a range of redshifts and projections. Noise has also been added to simulate a range of surface-brightness limits.

Postage stamps are all generated with a physical size of ~100 kpc × 100 kpc with a pixel scale of 0.2″ matching LSSTCam. For each classification you will inspect 4 images: the first 3 are greyscale log scaled images with different maximum stretches (27, 29 and 33 mag / sq. arcsec) and the final image is an arcsinh stretched *g,r,i* false colour image.

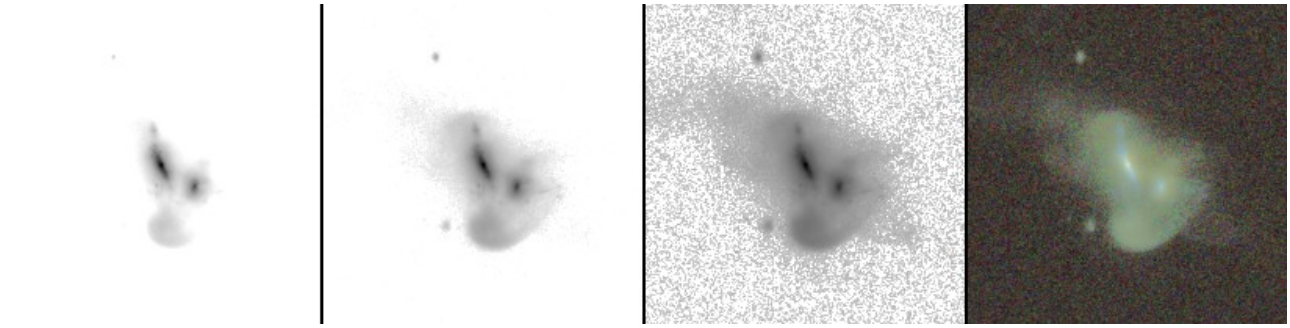

Figure 1: From left to right: greyscale image with log scaling, maximum stretch 27 mag / sq. arcsec, with a maximum stretch of 29 mag / sq. arcsec and maximum stretch of 33 mag / sq. arcsec, finally an arcsinh stretched *g,r,i* false colour image

Most images will be noisier and have lower spatial resolution that the ones above. This will effect the number of tidal features you are able to classify.

Classifications: Over the next 2 pages you will find descriptions and examples of the different categories that you will be asked to classify. You will note down the **number of instances** of each category of tidal feature in each object. Classifications should be in relation to the host galaxy (the object in the centre of the image) so **any features that are either clearly not associated with the host galaxy or not associated with a galaxy that is interacting with the host galaxy should be ignored**. At higher redshifts it is likely that that very few tidal features will be visible, but it may still be possible to identify ongoing mergers.

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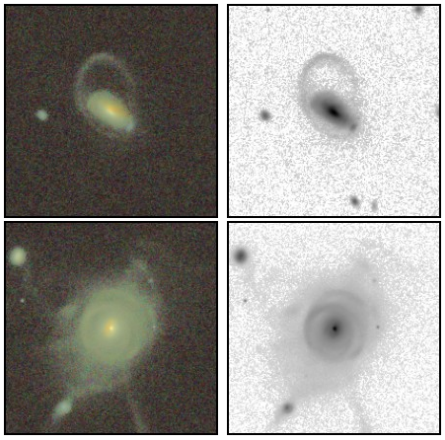
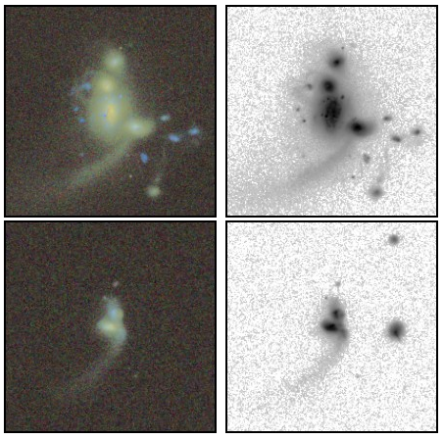
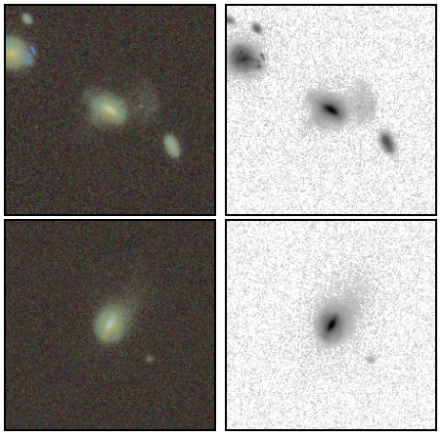
Stellar streams: A stream of stars orbiting the host galaxy. These can appear to trace an ellipse, spiral or straight line depending on the angle at which they are viewed. Note each separate instance: for example the top image has only one clear stellar stream, whereas the bottom one contains multiple examples.



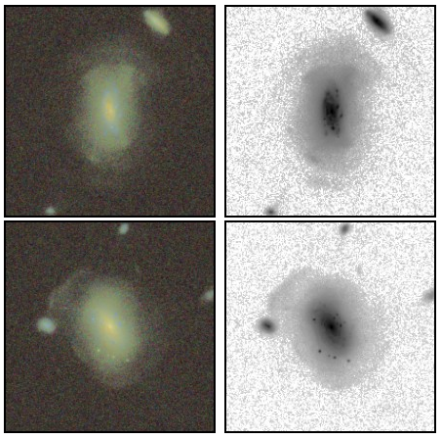
Tidal tails: Our definition of tidal tail differs from stellar streams in that a tidal tail must originate from material ejected from the host galaxy or a massive companion. They are therefore likely to be associated with interactions between similar mass galaxies and consist of material that has been unbound from a disrupted galaxy rather than gradually stripped.



Plumes or asymmetric stellar haloes: Diffuse, low-surface-brightness features in the outskirts of the galaxy that do not have a well defined structure like stellar streams. This also includes objects with miscellaneous LSB features, for example, where the overall structure of the diffuse stellar halo is clearly not symmetric.



Shells: Galaxies with radial structures consisting of concentric arcs or ring-like structures that do not cross one another. **You should note the total number of individual shells you see in the image.**



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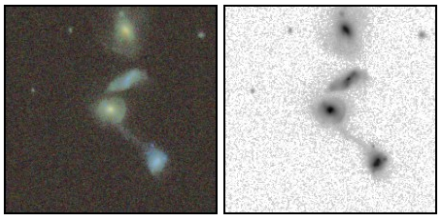
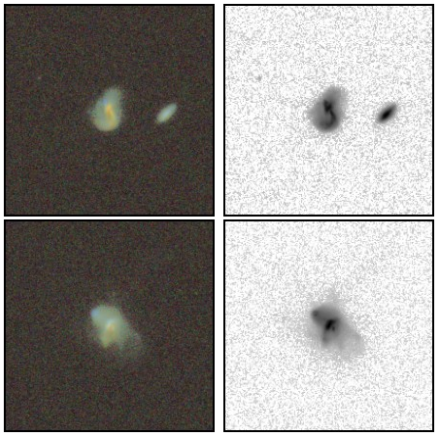
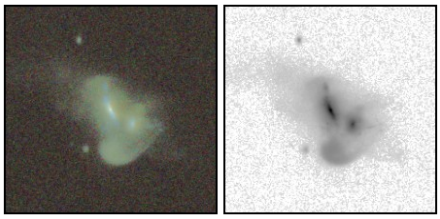
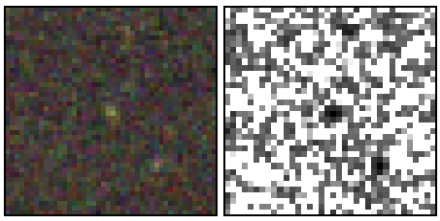
Tidal bridges: A bridge is a tidal tail that connects two interacting galaxies. Bridges are formed from the material that gets dragged out during high mass ratio mergers, rather than material has been gradually stripped away over many orbits.



Merger remnant: A strongly morphologically disturbed galaxy where the merging objects have recently coalesced. May also exhibit tidal tails but there should be no indication of a second galaxy (i.e. not pre-coalescence).



Double nuclei: Galaxy that is visibly merging / interacting, but where both objects are still clearly separated. **Do not count objects where there is no sign of an interaction taking place e.g. close pairs.**



Not detected or too noisy: In some cases the image will not be deep enough to reasonably detect any LSB features, or there may be no object detected at all. In this case you can just skip the classification.